MATH INQUIRERS HOLIDAY PROGRAM

Question Bank

Start here, Think harder and Go further questions for every lesson, with answers

Notice it. Test it. Explain why.

Question Bank • Level 4 (ages 9-11, covers P4 topics)

Math Inquirers Level 4 (ages 9-11). Notice it. Test it. Explain why.

How to use this bank

- Each lesson has three levels. Every child starts with Start here, then moves up when they are ready. Go further questions are the ones to talk about together.
- CORE questions can be started by a child who has not yet learnt the topic. They can use objects, drawing, counting or guess and check. Use them for warm-ups and to build confidence.
- THINK HARDER uses the idea taught in the lesson in a new setting: two-step problems, patterns with a hidden rule, unknowns and larger numbers.
- GO FURTHER questions are for children who have completed the topic. They need the topic plus an extension: find every way, work backwards, test a case, justify an answer, change the problem or make up your own. This program is inspired by research-based approaches used in gifted education; it is not an official GEP standard and is not affiliated with MOE.
- Ask for the reason every time: "How do you know?" A correct answer with no reason is only half done.
- Answers are at the back. Do not hand the answer pages to children.

What each lesson practises

Lesson 1	Factor Hunters (Factors and multiples)	6 + 8 + 9 questions
Lesson 2	Big Number Puzzles (Numbers to 100 000)	6 + 8 + 9 questions
Lesson 3	Operation Detectives (Four operations)	6 + 11 + 9 questions
Lesson 4	Fraction Frontier (Fractions)	6 + 9 + 9 questions
Lesson 5	Decimal Detectives (Decimals)	6 + 11 + 9 questions
Lesson 6	Angle Turns (Angles)	6 + 9 + 9 questions
Lesson 7	Area Architects II (Area and Perimeter)	6 + 9 + 9 questions
Lesson 8	Symmetry and Nets (Symmetry and Nets)	6 + 8 + 9 questions
Lesson 9	Data Detectives (Tables and line graphs)	6 + 8 + 9 questions
Lesson 10	Challenge Day (Mixed heuristics)	6 + 8 + 9 questions

Lesson 1: Factor Hunters

Factors and multiples • Week 1

START HERE

C1. Ali has 12 sweets. He puts them into bags so that every bag has the same number of sweets and none are left over. Use 12 counters, or draw 12 dots. Write down every number of sweets that one bag could have.

C2. A frog starts at 0 and jumps along a number line in jumps of 7. Write the first five numbers it lands on.

C3. Can 52 stickers be shared equally among 4 children with none left over? Try it: give each child 10 stickers first, then see how many are left to share.

C4. Mei counts 3, 6, 9, 12 … (in 3s). Ravi counts 5, 10, 15, 20 … (in 5s). Write a number less than 20 that both of them say.

C5. Make rectangles from 20 square tiles, using all of them each time. Draw every different rectangle you can on squared paper (a 2 by 10 and a 10 by 2 are the same). Write each one as rows × tiles in a row.

C6. A shop gives a free sticker with every 3rd purchase and a free pen with every 4th purchase. Write the numbers 1 to 12. Mark the purchases that get a sticker, then the ones that get a pen. Which purchase gets both?

THINK HARDER

T1. How many factors does 72 have? List them in order.

T2. One lighthouse flashes every 6 seconds. Another flashes every 9 seconds. They flash together now. After how many seconds will they next flash together?

T3. Find all the common factors of 24 and 36.

T4. Find ALL the numbers from 1 to 60 that have exactly 3 factors. What do the numbers have in common?

T5. Ben shares 28 stickers equally among some friends with none left over. He has more than 1 friend and gives each friend more than 1 sticker. How many friends could he have? Find all the answers.

T6. I am a multiple of 7 between 50 and 80. My digits add up to 11. Who am I?

T7. Which number less than 30 has the most factors?

T8. Mrs Tan has 36 pencils and 48 erasers. She puts them into identical packs with nothing left over. What is the greatest number of packs she can make? What is in each pack?

GO FURTHER (non-routine: justify, find every way)

X1. How many numbers from 1 to 100 are multiples of 4 or multiples of 6 (or both)?

X2. Mei has between 30 and 50 eggs. When she packs them in 4s, 1 egg is left. When she packs them in 5s, 1 egg is left. How many eggs does she have?

X3. What is the smallest number that has exactly 6 factors?

X4. Ali counts in 6s from 6. Mei counts in 8s from 8. Both stop at 100. How many numbers do they both say?

X5. Ben says: "A number that is a multiple of 4 and a multiple of 6 must be a multiple of 24, because 4 × 6 = 24." Test his idea with examples. Is he right? What is true instead?

X6. Find a number between 20 and 40 that has exactly 4 factors and is odd.

X7. Bus A leaves a stop every 6 minutes and Bus B every 9 minutes. Bus C leaves every ___ minutes. All three leave together at 9:00 am, and they next leave together at 9:36 am. What could the gap for Bus C be? Find ALL the whole-minute answers. Then make up a puzzle like this one for a partner.

X8. I am a number less than 100. When I am divided by 2, 3, 4, 5 or 6, there is always 1 left over. I am bigger than 1. Who am I?

X9. There are 20 closed lockers, numbered 1 to 20. Child 1 opens every locker. Child 2 changes every 2nd locker (opens it if it is closed, closes it if it is open). Child 3 changes every 3rd locker, and so on, up to child 20. Which lockers are open at the end?

Lesson 2: Big Number Puzzles

Numbers to 100 000 • Week 1

START HERE

C1. In 64 205, what is the digit 4 worth?

C2. Write fifty-three thousand and eighty in numerals.

C3. Mei's number is 27 649. Is it closer to 27 000 or to 28 000? Draw a number line from 27 000 to 28 000 and mark the halfway point (27 500) to help you.

C4. What number is 1000 less than 70 230?

C5. Put these numbers in order, smallest first: 45 099, 45 909, 45 990, 45 900.

C6. A bakery baked 6352 buns. Is that closer to 6300 or to 6400? Draw a number line from 6300 to 6400 and mark the halfway point (6350).

THINK HARDER

T1. I am a 5-digit number. All my digits are the same. Rounded to the nearest ten thousand, I am 60 000. Who am I?

T2. Use the digit cards 1, 4, 6, 7 and 9 once each. Make the 5-digit number that is closest to 70 000.

T3. Find the rule and write the next two numbers: 17 250, 19 500, 21 750, ____, ____

T4. How many whole numbers round to 3500 when rounded to the nearest hundred?

T5. A whole number rounds to 5350 to the nearest 10, AND to 5400 to the nearest 100. Find ALL the numbers it could be.

T6. Estimate 8 × 6102 by rounding 6102 to the nearest thousand first. Is the exact answer more or less than your estimate?

T7. The crowd at the first concert was 38 000 to the nearest thousand. At the second concert it was 41 000 to the nearest thousand. What is the smallest and the largest the difference between the two crowds could be?

T8. What is the largest 5-digit odd number with all its digits different?

GO FURTHER (non-routine: justify, find every way)

X1. Ali counts on in 1000s, starting at 3456. He stops at the last number below 100 000. What is the last number he says? How many numbers does he say?

X2. How many whole numbers round to 45 000 to the nearest thousand AND to 50 000 to the nearest ten thousand?

X3. A palindrome reads the same forwards and backwards, like 12 321. How many 5-digit palindromes are there between 40 000 and 50 000?

X4. A 5-digit number has all its digits different, and its digits add up to 10. What is the largest it can be? What is the smallest?

X5. Find all the 5-digit palindromes that round to 50 000 when rounded to the nearest thousand.

X6. Ben writes all the numbers from 99 990 to 100 000. How many digits does he write altogether?

X7. How many 5-digit numbers have digits that add up to 3?

X8. Make up your own number riddle about a 5-digit number. Use a rounding clue, a clue about the digits and an odd-or-even clue, so that exactly ONE number fits. Check your riddle: list every number that fits the rounding clue, then cross out the ones that break the other clues. Swap riddles with a partner.

X9. The distance counter on Mr Tan's car shows 39 993 km. This number is a palindrome: it reads the same forwards and backwards. What is the next palindrome the counter will show? How many more kilometres is that?

Lesson 3: Operation Detectives

Four operations • Week 2

START HERE

C1. Ali has 18 stickers. He buys 6 packs with 3 stickers in each pack. How many stickers does he have now? Then write ONE number sentence that shows what you did.

C2. 18 boys and 6 girls each buy 3 tickets. How many tickets are bought altogether? Then write ONE number sentence for it.

C3. A ribbon 56 cm long is cut into 7 equal pieces. Ben takes 2 of the pieces. How long is Ben's ribbon?

C4. At a camp there are 90 children. 30 of them make 5 equal teams for a game. After the game, one whole team goes home. How many children are left at the camp?

C5. Work out 2 345 × 3.

C6. Work out 4 236 ÷ 6.

THINK HARDER

T1. There are 2 340 red and blue beads. There are 4 times as many blue beads as red beads. How many red beads are there?

T2. Put ONE pair of brackets in 3 + 5 × 6 − 2 to make the answer 46.

T3. Write +, −, × or ÷ in each gap to make this true: 12 __ 4 __ 5 = 15

T4. Think of a number. Multiply it by 5. Add 35. Divide by 5. Take away the number you started with. The answer is always 7. Explain why it always works. A bar model can help.

T5. Work out 120 − 4 × (18 + 7).

T6. In this chain, each number after the first two is the sum of the two before it. Fill in the gaps: 3, __, __, 17, __

T7. Use each of the numbers 2, 3, 4 and 5 once in a + b × c − d (a, b, c and d are the four numbers, in any order) to make the biggest answer you can.

T8. Is 50 − 20 − 10 the same as 50 − (20 − 10)? Explain.

T9. I think of a number. I multiply it by 8 and then take away 15. The answer is 425. What is my number?

T10. Look at 4 + 8 × 5 − 3. Mei puts TWO pairs of brackets in it and makes 24. Where could she put them?

T11. A hawker stall sold 3 times as many plates of chicken rice as plates of laksa. It sold 1 236 plates altogether. How many more plates of chicken rice than laksa did it sell? (Draw a bar model first.)

GO FURTHER (non-routine: justify, find every way)

X1. Use four 4s and any of +, −, ×, ÷ and brackets to make each of the numbers 1, 2, 3, 4 and 5. Which one was hardest?

X2. Ali has $56 more than Ben. Together they have $300. How much money does Ben have?

X3. Use 3, 4, 5 and 6 once each, with any signs and brackets, to make 24. Find two different ways.

X4. The same number a is used three times in a × a − a = 30. What is the number a?

X5. In a number chain, each number after the first two is the sum of the two before it. The 5th number is 30 and the 6th number is 49. What is the 1st number?

X6. Ravi has 3 times as many stickers as Siti. Ravi gives 20 of his stickers to Siti, and now they have the same number. How many stickers did each child have at first?

X7. Put + or − in each gap so that the answer is 7: 1 __ 2 __ 3 __ 4 __ 5 __ 6 = 7. Find ALL the ways.

X8. Mei spent half of her money on a book. Then she spent $6 on lunch. Then she spent half of what was left on a gift. She had $9 left. How much money did she have at first?

X9. The number trick: think of a number, multiply by 5, add 35, divide by 5, take away your number. It always ends with 7. (a) What must you change so that it always ends with 10? (b) Make up your own number trick with four steps that always ends with 3. Test it with two numbers and explain why it works.

Lesson 4: Fraction Frontier

Fractions • Week 2

START HERE

C1. Ravi eats 11 slices of pizza. Each pizza is cut into 3 equal slices. Draw circles cut into 3 slices to help you. How many whole pizzas did he eat, and how many slices of another pizza?

C2. Mei has 2 whole chocolate bars and 3 more pieces. Each bar is broken into 4 equal pieces. Draw the bars. How many pieces does she have altogether?

C3. There are 32 counters. Put them equally into 4 groups. How many counters are in one group? Ali takes one group. What fraction of the counters does he take?

C4. Put 18 counters equally into 3 groups. Give 2 of the groups to Mei. How many counters does Mei get?

C5. Fold two paper strips of the same length. Fold one into 6 equal parts and shade 5 of them. Fold the other into 3 equal parts and shade 2 of them. Which strip has more shaded?

C6. A ribbon 1 m long is marked in 10 equal parts. Mei uses 3 parts for a bow and 4 more parts to wrap a gift. What fraction of the ribbon is left?

THINK HARDER

T1. Ben ate 7/4 pizzas and Ravi ate 1 5/8 pizzas. Who ate more?

T2. Siti has 40 beads. 3/8 of them are blue. How many beads are NOT blue?

T3. Put these fractions in order, smallest first: 2/3, 1/2, 5/6, 7/12

T4. Who am I? I am an improper fraction. My denominator is 6. I am between 2 and 3. My numerator is an odd number. I am closer to 3 than to 2. Make a list, then cross out the ones that do not fit.

T5. Find ALL the fractions with denominator 10 that are bigger than 1/5 but smaller than 1/2.

T6. A ribbon is 3 m long. Mei cuts it into pieces that are each 1/4 m long. How many pieces does she get?

T7. How many quarters are there in 4 3/4?

T8. Ravi says 3/7 is bigger than 3/5 because 7 is bigger than 5. Is he right? Explain.

T9. 2/5 of a number is 16. What is the number?

GO FURTHER (non-routine: justify, find every way)

X1. I am a fraction equal to 3/4. My numerator and denominator add up to 28. Who am I?

X2. Mei spent 1/3 of her money on a book and 1/4 of her money on a pen. She had $25 left. How much money did she have at first?

X3. A tank is 3/8 full of water. After 15 litres more are poured in, it is 3/4 full. How many litres does the full tank hold?

X4. Use two of the digit cards 2, 3, 5 and 7 to make a fraction smaller than 1 that is as close to 1 as possible.

X5. In a bag of marbles, 1/4 are red and 1/3 are blue. The other 10 marbles are green. How many marbles are in the bag?

X6. Ben gives 1/3 of his marbles to Ali and 1/4 of his marbles to Mei. Ali gets 2 more marbles than Mei. How many marbles did Ben have at first?

X7. Mei takes 1/2 of the sweets in a jar. Ali takes 1/3 of the sweets. There are 5 sweets left. (a) How many sweets were in the jar at first? (b) What if 7 sweets were left? (c) Make up your own puzzle with two other fractions taken from a jar and a number of sweets left, and solve it.

X8. Half of a number is 3 more than a third of the number. What is the number?

X9. Siti read 2/5 of her book on Monday. On Tuesday she read half of the pages that were left. She still has 45 pages to read. How many pages are in the book?

Lesson 5: Decimal Detectives

Decimals • Week 3

START HERE

C1. Ali has 3 one-dollar coins, 4 ten-cent coins and 7 one-cent coins. How much money is that? Write the amount in dollars, using a decimal point.

C2. Ali has $0.90 and Mei has $0.89. Who has more money? How much more? (Think in cents.)

C3. Is 7.46 closer to 7.4 or to 7.5? Draw a number line from 7.4 to 7.5 and mark the halfway point 7.45 to help you.

C4. Ali buys 3 pens at $1.25 each. How much does he pay? (Add $1.25 three times, or think of $1.25 as 125 cents.)

C5. A stick is 1 metre long. It is cut into 10 equal parts and Mei takes 4 parts. She then cuts one more part into 10 tiny equal pieces and takes 7 of them. How long is Mei's stick, written as a decimal of a metre?

C6. A number line from 3 to 4 has 10 equal steps. An arrow points to the 6th step after 3. What number is it? Now zoom in on the gap between that number and the next step, and cut it into 10 smaller steps. A new arrow points to the 5th small step. What number is that?

THINK HARDER

T1. Find ALL the digits that the letter A can be: 3.A5 < 3.42 (A is one digit)

T2. A number with 2 decimal places rounds to 10 when rounded to the nearest whole number. What is the smallest it can be? The largest?

T3. What number is halfway between 2.6 and 2.7? What number is halfway between 2.6 and 2.61?

T4. Mystery decimal. I have 3 decimal places and I am between 4 and 5. Rounded to 1 decimal place, I am 4.7. My hundredths digit is 2 more than my tenths digit. My digits add up to 25. Who am I?

T5. Siti buys 4 notebooks that cost the same. She pays with a $10 note and gets $2.60 change. How much does one notebook cost?

T6. Write these in order, smallest first: 0.6, 0.066, 0.606, 0.66, 0.06

T7. Mei is 1.38 m tall. Ben is 0.15 m taller than Mei. Ali is 0.09 m shorter than Ben. How tall is Ali?

T8. Use the digit cards 2, 4, 7 and a decimal point once each (the point goes between two digits). Which number is closest to 50?

T9. A ribbon 4.8 m long is cut into 6 equal pieces. How long is each piece? Ravi uses 4 of the pieces. How much ribbon does he use?

T10. A number has 2 decimal places. Rounded to 1 decimal place it is 7.4. How many different numbers could it be? Its digits add up to 18. What is the number?

T11. Use the digit cards 3, 6, 9 and a decimal point card once each. The point can go between any two digits (like 3.69 or 36.9). How many different numbers can you make? Which is closest to 7? Which rounds to 40 to the nearest whole number?

GO FURTHER (non-routine: justify, find every way)

X1. A number with 3 decimal places rounds to 2.46 when rounded to 2 decimal places. Its digits add up to 20. What is the number?

X2. 3 apples and 1 pear weigh 0.95 kg. 1 apple and 1 pear weigh 0.45 kg. All the apples weigh the same. What is the mass of the pear?

X3. Ravi has 10 coins. They are all 20¢ and 50¢ coins. Altogether they make $3.80. How many of each coin does he have?

X4. Ben rounds a number with 2 decimal places to the nearest whole number and gets 6. Mei rounds the same number to 1 decimal place and gets 5.5. Find ALL the numbers it could be.

X5. A pen costs $0.85 more than a pencil. 2 pens and 1 pencil cost $4.70 altogether. How much does a pencil cost?

X6. How many numbers with exactly 2 decimal places are between 3.9 and 4.1? (Do not count 3.9 and 4.1.)

X7. Make up your own "closest to" puzzle. Choose four digit cards and a decimal point (the point goes between two digits). Pick a target number and find the number with 2 decimal places that is closest to it. Then ask: what if the target changes? Swap puzzles with a partner and check each other's answers.

X8. Ali, Ben and Siti share $9.20. Ben gets $0.50 more than Ali. Siti gets $1.30 more than Ben. How much does each child get?

X9. A jug full of water weighs 1.45 kg. When it is half full, it weighs 0.85 kg. What is the mass of the empty jug?

Lesson 6: Angle Turns

Angles • Week 3

START HERE

C1. Stand up and face the front. Turn all the way round until you face the front again: that is a full turn, 360°. How many degrees is a half turn? How many degrees is a quarter turn?

C2. The corner of a page is a right angle, 90°. How many right angles make a full turn? How many degrees is a turn of 3 right angles (3/4 of a turn)?

C3. Ali faces East. He turns a quarter turn clockwise (the way the clock hands go). Which direction does he face now? Stand up and try it.

C4. A compass has 8 directions: N, NE, E, SE, S, SW, W, NW. Each step to the next one is 45°. Mei faces N and turns 3 steps anticlockwise. Which direction does she face now? How many degrees did she turn?

C5. A clock has 12 equal gaps between the numbers, all the way round the 360° circle. How many degrees is one gap? What is the angle between the hands at 1 o'clock? At 6 o'clock?

C6. The corner of a rectangle is 90°. A line from the corner cuts it into two angles. One angle is 64°. What is the other?

THINK HARDER

T1. Siti faces SE. She turns 225° clockwise. Which direction does she face now?

T2. What is the angle between the clock hands at 10 o'clock? At 2 o'clock? What do you notice?

T3. How many degrees does the minute hand turn from 7:15 to 7:50?

T4. A robot starts facing N. Ben wants it to turn from N to W using two DIFFERENT cards from: 45° clockwise, 90° clockwise, 135° anticlockwise, 180°. Find ALL the pairs that work.

T5. Mei faces W. She turns clockwise until she faces SE. How many degrees does she turn?

T6. Four rays come out of one point, making 3 angles side by side: 25°, 40° and 25°. How many angles are there altogether? What is the biggest?

T7. Ravi faces N and turns anticlockwise to face S. Then he turns anticlockwise again to face E. How many degrees has he turned altogether?

T8. The minute hand of a clock turns 150° starting at 2:10. What time is it then?

T9. The corner of a square is cut into two angles. One angle is 4 times as big as the other. Find both angles.

GO FURTHER (non-routine: justify, find every way)

X1. What is the angle between the hands of a clock at 3:30?

X2. Five rays come out of one point and make 4 equal angles side by side. The angle between the two outside rays is 120°. How many angles can you find altogether? How many of them are right angles?

X3. A robot faces N. It turns 45° clockwise, again and again. After how many turns does it first face N again? Which way does it face after 30 turns?

X4. At 4 o'clock the angle between the clock hands is 120°. What is the angle at 4:10?

X5. Mei says: "At 2 o'clock the angle between the clock hands is 60°, so at 10 o'clock it must be 60° too." Is she right? Explain with a drawing. Which other pairs of o'clock times have the same angle?

X6. Mei faces S and turns clockwise to face NE. Ali faces S and turns anticlockwise to face NE. How many more degrees did Mei turn than Ali?

X7. Make up your own clock-angle puzzle. Choose a time that is NOT an o'clock time (half past is easiest), work out the angle between the hands, and write your puzzle as "What is the angle at ...?". Draw the clock to check your answer. Then find a second time with the same angle.

X8. The corner of a square is cut into 3 angles. The middle angle is twice the smallest. The biggest is 3 times the smallest. Find all three angles.

X9. Two squares share one corner O and overlap, like the picture. The angle covered by both squares together (from the outside side of one square to the outside side of the other) is 160°. What is the angle where the squares overlap?

Lesson 7: Area Architects II

Area and Perimeter • Week 4

START HERE

C1. Draw a rectangle 9 cm long and 7 cm wide on squared paper (1 cm squares). How many squares are inside it? That is its area in cm².

C2. A rectangle is 15 cm long and 4 cm wide. Walk your finger once round the edge, adding the sides as you go. How long is the way round (the perimeter)?

C3. Use 12 square tiles (each 1 cm by 1 cm). Make a rectangle using all of the tiles. Write its length and breadth. Can you make a different rectangle with the 12 tiles? Draw each one.

C4. A rectangle has an area of 56 cm². Its length is 8 cm. Find its perimeter. (Draw 8 squares in a row and add rows until you have 56 squares.)

C5. A figure is made of a 6 cm by 4 cm rectangle and a 3 cm by 2 cm rectangle joined together without overlapping. Draw it on squared paper. Find its area.

C6. On squared paper draw a 6 cm by 6 cm square. Cut a 2 cm by 2 cm square from one corner. How many squares are left (the area)? Walk round the edge of the new shape: is the perimeter more than, less than, or the same as the perimeter of the 6 cm by 6 cm square?

THINK HARDER

T1. List every rectangle with whole-cm sides and an area of 30 cm². Which one has the smallest perimeter?

T2. A rectangle has whole-cm sides and a perimeter of 20 cm. List all the possible rectangles and their areas. Which has the largest area?

T3. A 3 cm by 3 cm square is cut from each of the four corners of a 12 cm by 8 cm rectangle. Find the area and the perimeter of the shape that is left.

T4. Three squares with sides 2 cm, 3 cm and 4 cm stand side by side in a row on the same line, smallest on the left. Find the area and the perimeter of the figure.

T5. A path 2 m wide goes all round a lawn 10 m long and 6 m wide. Find the area of the path.

T6. Ali joins 8 square tiles, each 1 cm by 1 cm, edge to edge. What is the smallest perimeter his shape can have? The largest?

T7. A 6 cm by 4 cm rectangle is cut into two 6 cm by 2 cm rectangles, which are put end to end to make a 12 cm by 2 cm rectangle. How much longer is the new perimeter?

T8. A rectangle has whole-cm sides, an area of 40 cm² and a perimeter of 26 cm. Find its length and breadth.

T9. Which rectangle with whole-cm sides and an area of 36 cm² has the smallest perimeter? What is its perimeter?

GO FURTHER (non-routine: justify, find every way)

X1. A square is cut into two identical rectangles. Each rectangle has a perimeter of 24 cm. Find the area of the square.

X2. A gap is cut into the middle of the top side of a 10 cm by 6 cm rectangle. The gap is 2 cm wide and 4 cm deep, with square corners. Find the area and the perimeter of the shape that is left.

X3. Ravi has 24 m of fencing to make a rectangular pen with whole-metre sides. (a) What is the largest area he can fence? (b) Now one side of the pen is a wall and needs no fence. What is the largest area now?

X4. Two 8 cm by 2 cm rectangles cross at their middles to make a plus shape. Find the area and the perimeter of the plus shape.

X5. Siti's garden is a square with an area of 49 m². She makes it 1 m longer and 1 m wider. How much bigger is the area? Then try 2 m longer and wider, and 3 m. Write what you notice.

X6. Mei makes a rectangle with 36 square tiles (each 1 cm by 1 cm). Ben makes a different rectangle with 36 tiles. Ben's perimeter is 16 cm longer than Mei's. What are the two rectangles?

X7. Make up your own fencing puzzle like Ravi's: choose a length of fence (a whole number of metres, more than 20) and say whether one side is a wall. Find the largest area the pen can have, and prove it by listing every whole-metre pen. Swap puzzles with a partner and check each other's lists.

X8. A garden is 10 m long and 8 m wide. Two paths 1 m wide cross it: one goes along its length and one goes across its width. What area of the garden is NOT path?

X9. A rectangle 9 cm long and 4 cm wide is cut into 2 pieces. The pieces fit together exactly to make a square. How long is a side of the square? Did the perimeter go up or down?

Lesson 8: Symmetry and Nets

Symmetry and Nets • Week 4

START HERE

C1. Hold a box or a dice. How many faces, edges and corners does a cube have? Count them one by one.

C2. Fold a square piece of paper so that the two halves match exactly. How many different ways can you fold it like this? (Folds from corner to corner count too.)

C3. Now fold a rectangle that is not a square so that the two halves match exactly. How many different ways can you fold it?

C4. Look at a dice. Opposite faces add up to 7. Which number is on the face opposite 2?

C5. Write the capital letters A, B, F, H, L and T on paper in big block letters. Which of them can be folded so that the two halves match exactly?

C6. Tape 6 squares in a row. Can you fold the row into a cube with no gaps and no overlaps? What goes wrong?

THINK HARDER

T1. A cuboid is 3 cm long, 2 cm wide and 1 cm high. Its net is drawn on 1 cm squared paper. How many squares does the net cover?

T2. Ben puts a dice on the table. The top face shows 3 and the front face shows 5. What numbers are on the bottom face and on the back face?

T3. A dice is on a table. Mei walks round it and can see the top and the four side faces. The numbers she can see add up to 19. What number is on the bottom face?

T4. There are 5 different shapes made of 4 squares joined edge to edge (a line, a square, an L, a T and a zigzag). Which one has exactly 1 line of symmetry?

T5. Can a 2 by 3 rectangle of squares fold into a cube? Explain.

T6. Which shape made of 5 squares joined edge to edge has exactly 2 lines of symmetry?

T7. Ben folds a net into a cube. Each face is a different colour. How many faces can touch the red face along an edge? Could 5 faces touch it?

T8. On a 3 by 3 grid of squares, shade 3 squares so that the pattern has exactly 1 line of symmetry. Then shade 2 squares so that the pattern has exactly 2 lines of symmetry. Draw both patterns.

GO FURTHER (non-routine: justify, find every way)

X1. Two dice are glued together face to face, and the two glued faces show the same number. The numbers on all the faces you can still see add up to 36. What number is on the glued faces?

X2. Show why a net of a cube can never have 5 squares in a row. What is the most squares in a row a net can have?

X3. An open box (a cube with no lid) is made from 5 squares. Of the 12 different shapes made of 5 squares joined edge to edge, how many fold into an open box? (Turned or flipped copies count as the same.)

X4. Three dice are stacked in a tower on a table. The top face of the tower shows 5. What is the total of the numbers on all the faces you cannot see (the faces touching each other and the table)?

X5. Ali says: "On a dice, the numbers on the 4 side faces always add up to 14." Is he right? Explain.

X6. A cube has faces numbered 1 to 6 (not a normal dice). Face 1 touches faces 2, 3, 4 and 5. Face 2 touches faces 1, 3, 5 and 6. Which face is opposite 1? Opposite 2? Opposite 3?

X7. Design your own dice. Draw a net of a cube that is not one of the nets in this lesson, and write the numbers 1 to 6 on it so that opposite faces add up to 7. Name each pair of opposite faces to prove it works. Then rub out two numbers and challenge a partner to put them back.

X8. In how many ways can you shade exactly 2 squares of a 3 by 3 grid so that the up-down line through the middle is a line of symmetry?

X9. Shade exactly 5 squares of a 3 by 3 grid so that the pattern has 4 lines of symmetry. There are 2 ways. Find both.

Lesson 9: Data Detectives

Tables and line graphs • Week 5

START HERE

C1. On a line graph, each line on the scale is worth 10. Draw the scale from 30 to 40. A point is halfway between the 30 line and the 40 line. What number does it show?

C2. A stall sold 45 bowls of laksa on Monday and 60 bowls on Tuesday. By how many bowls did sales go up?

C3. A table shows 32 visitors on Monday, 48 on Tuesday and 20 on Wednesday. How many visitors came altogether?

C4. A line graph goes up from 8 am to 10 am, then down until 12 pm. Sketch what it looks like. At what time was the number the highest?

C5. A table shows that Ali has 25 cards and Mei has 18 more cards than Ali. Draw a bar for each child. How many cards do they have altogether?

C6. Three days of sales add up to 90. Day 1 is 30 and Day 2 is 25. What is Day 3?

THINK HARDER

T1. A table shows that Class 4A has 28 boys and 6 more girls than boys. Class 4B has 63 pupils, with 5 more girls than boys. Which class has more girls?

T2. On a line graph each line on the scale is worth 20. A point is 3 lines above the 40 line. What number does it show?

T3. A shop sold 300 kites in 4 days. Day 1: 65, Day 2: 80, Day 4: 90. How many kites were sold on Day 3?

T4. Ali's height: age 7, 120 cm; age 8, 127 cm; age 9, 131 cm; age 10, 137 cm. In which year did he grow the most? How much did he grow altogether from age 7 to age 10?

T5. Shop A sold 30, 35, 40 and 45 cakes in weeks 1 to 4. Shop B sold 50, 48, 46 and 44 cakes. In which week did Shop A first sell more than Shop B?

T6. A graph's scale starts at 40. Mei says a point at 60 is "double" a point at 50 because it is twice as high on the graph. Is she right? Explain.

T7. Visitors to a museum: Monday 120, Tuesday 95, Wednesday 140, Thursday 110. Between which two days was the biggest change? Was it an increase or a decrease?

T8. The temperature went up by 2 °C every hour from 8 am to 1 pm. At 1 pm it was 33 °C. What was the temperature at 8 am?

GO FURTHER (non-routine: justify, find every way)

X1. A shop's sales go up by the same amount every week. In week 2 it sold 45 toys and in week 6 it sold 85 toys. How many toys did it sell in week 1? In which week did it first sell more than 100 toys?

X2. A stall sold 250 bowls of noodles from Monday to Friday. Monday 40 and Friday 50. Tuesday and Wednesday were the same, and Thursday was 10 more than Tuesday. Find the sales for Tuesday, Wednesday and Thursday.

X3. On Ben's graph each line is worth 2. On Mei's graph each line is worth 10. Both lines go up by 3 grid lines from Monday to Tuesday. Whose number went up more, and by how much more?

X4. A graph's scale starts at 200 and each line is worth 50. Monday's point is 2 lines up and Tuesday's point is 6 lines up. Ravi says Tuesday's number is 3 times Monday's. Is he right? Explain.

X5. At a pool, the number of swimmers went up by 15 every hour from 9 am to 11 am, then went down by 10 every hour until 1 pm. At 1 pm there were 50 swimmers. How many were there at 9 am?

X6. A stall sold 400 plates of nasi lemak from Monday to Friday. Each day it sold 10 more plates than the day before. How many plates did it sell on Monday?

X7. Four teams scored 100 points altogether. Team A scored twice as many as Team B. Team C scored 10 more than Team B. Team D scored 22. How many points did each team score?

X8. Tank A has 80 litres of water and loses 6 litres every hour. Tank B has 20 litres and gains 9 litres every hour. On a line graph the two lines cross. After how many hours do the tanks hold the same amount? How much is in each?

X9. Four numbers: 52, 54, 56, 58. Choose where the scale starts and how much each line is worth, so that a line graph of these numbers looks (a) like a huge increase and (b) like hardly any increase. Write both scales and a sentence a shop owner might say with each graph. Which graph is fair? Why?

Lesson 10: Challenge Day

Mixed heuristics • Week 5

START HERE

C1. Two next-door numbers (like 7 and 8) multiply to make 90. What are they? Try some pairs and write each try in a list.

C2. A farm has chickens and rabbits. There are 8 heads and 22 legs. How many rabbits are there? Draw 8 animals with 2 legs each first, then see how many legs you still need to add.

C3. I think of a number. I add 15, then double it. I get 70. What is my number? Start at 70 and undo each step.

C4. Ali is 6th from the front and 6th from the back of a queue. How many people are in the queue? Draw the queue.

C5. Two numbers add up to 30. One is 6 more than the other. What are they?

C6. Ben has 6 coins. Some are 10¢ and some are 50¢. They make $1.40 altogether. How many 50¢ coins does he have? Try 1 fifty-cent coin, then 2, then 3 …

THINK HARDER

T1. There are 20 bicycles and tricycles with 47 wheels altogether. How many tricycles are there?

T2. Two next-door even numbers (like 6 and 8) multiply to make 624. What are they?

T3. Ravi spent $15 on a book. Then he spent half of the money left on a bag. He had $18 left. How much money did he have at first?

T4. Ali, Ben and Siti are 9, 10 and 11 years old. Ben is older than Siti. Siti is not the youngest. How old is each child?

T5. Mei is 9th from the front and 14th from the back of a queue. How many people are in the queue?

T6. There are 12 spiders (8 legs) and ants (6 legs) with 82 legs altogether. How many spiders are there?

T7. Three next-door numbers add up to 117. What is the biggest of the three?

T8. Siti and her dad are 48 years old altogether. In 6 years, her dad will be 4 times as old as Siti. How old is Siti now?

GO FURTHER (non-routine: justify, find every way)

X1. How many whole numbers from 1 to 100 have the digit 7 in them at least once?

X2. A farm has 20 chickens and goats. The chickens have 4 more legs altogether than the goats. How many chickens are there?

X3. I am a 2-digit number. My digits add up to 11. If you swap my digits, the new number is 27 more than me. Who am I?

X4. Ali, Ben, Mei and Siti sit in a row of 4 seats, numbered 1 to 4 from the left. Mei sits at one end. Ben sits next to Mei. Ali does not sit next to Ben. Ali sits in seat 1 or seat 2. Who sits in each seat?

X5. A cake costs $3 more than a drink. 4 cakes and 3 drinks cost $40. How much does a drink cost?

X6. Ben's uncle is 4 times as old as Ben. In 10 years, his uncle will be only twice as old as Ben. How old is Ben now?

X7. A hawker centre has 15 tables. Some seat 4 people and some seat 6 people. There are 76 seats altogether. How many tables seat 6?

X8. Mei gave Ali half of her stickers and 2 more. Then she gave Ben half of the stickers left and 2 more. She had 5 stickers left. How many did she have at first?

X9. Make up your own heads-and-legs puzzle (chickens and goats, or bicycles and cars) that has exactly ONE answer. Check it by trying every number of one animal. Then explain why a farmer who counts 12 heads and 35 legs on a farm with only chickens and goats must have counted wrong.

Answer key (tutor only)

Lesson 1: Factor Hunters

C1. 1, 2, 3, 4, 6 or 12 sweets in a bag.

C2. 7, 14, 21, 28, 35

C3. Yes. 40 stickers are given out first and 12 are left. 12 shared among 4 is 3 each, so each child gets 13 (4 × 13 = 52).

C4. 15

C5. 1 × 20, 2 × 10, 4 × 5

C6. The 12th purchase (stickers: 3, 6, 9, 12; pens: 4, 8, 12).

T1. 12 factors: 1, 2, 3, 4, 6, 8, 9, 12, 18, 24, 36, 72.

T2. 18 seconds (6, 12, 18 and 9, 18).

T3. 1, 2, 3, 4, 6, 12

T4. 4, 9, 25 and 49. Each is a number times itself (2 × 2, 3 × 3, 5 × 5, 7 × 7), where that number has only 2 factors. 16 = 4 × 4 has 5 factors and 36 has 9.

T5. 2, 4, 7 or 14 friends.

T6. 56 (the multiples of 7 are 56, 63, 70, 77; only 5 + 6 = 11).

T7. 24, with 8 factors: 1, 2, 3, 4, 6, 8, 12, 24.

T8. 12 packs, each with 3 pencils and 4 erasers.

X1. 33. Multiples of 4: 25. Multiples of 6: 16. Both (multiples of 12): 8, counted twice. 25 + 16 − 8 = 33.

X2. 41. One less than the number of eggs is a common multiple of 4 and 5 (20, 40, 60), so 21, 41 or 61; only 41 is between 30 and 50.

X3. 12 (1, 2, 3, 4, 6, 12). Every number below 12 has fewer than 6 factors.

X4. 4: 24, 48, 72, 96.

X5. Ben is wrong. 12 is a multiple of both 4 and 6, but it is not a multiple of 24. The common multiples of 4 and 6 are 12, 24, 36, 48 ...: every one is a multiple of 12, the smallest common multiple. 24 is only the second.

X6. 21 (1 × 21, 3 × 7), 27 (1 × 27, 3 × 9), 33 (1 × 33, 3 × 11), 35 (1 × 35, 5 × 7) or 39 (1 × 39, 3 × 13). Any one of these.

X7. 4, 12 or 36 minutes. The smallest common multiple of 6 and 9 is 18, so Bus C's gap must be a factor of 36 that makes the first meeting 36 (not 18). 1, 2, 3, 6, 9 and 18 all give 18, so they do not work. Puzzles will vary: check that the partner's answer really gives the stated time.

X8. 61. One less than the number is a common multiple of 2, 3, 4, 5 and 6: 60. 60 + 1 = 61 (the next one, 121, is too big).

X9. 1, 4, 9 and 16. Each locker is changed once for every factor of its number. Only numbers with an odd number of factors (1, 4, 9, 16: a number times itself) end up open.

Lesson 2: Big Number Puzzles

C1. 4000 (4 thousands)

C2. 53 080

C3. 28 000 (27 649 is past the halfway point 27 500).

C4. 69 230

C5. 45 099, 45 900, 45 909, 45 990

C6. 6400 (6352 is just past the halfway point 6350).

T1. 55 555 (66 666 rounds to 70 000; 44 444 rounds to 40 000).

T2. 69 741 (259 away; the closest number above 70 000 is 71 469, which is 1469 away).

T3. Add 2250 each time: 24 000, 26 250.

T4. 100 (from 3450 to 3549).

T5. 5350, 5351, 5352, 5353, 5354 (5 numbers). To the nearest 10 it is 5345 to 5354; to the nearest 100 it is 5350 to 5449; both: 5350 to 5354.

T6. 8 × 6000 = 48 000. More, because 6102 was rounded down (exact: 48 816).

T7. Smallest 2001 (40 500 − 38 499). Largest 3999 (41 499 − 37 500).

T8. 98 765

X1. 99 456. He says 97 numbers (3456, 4456, ... 99 456).

X2. 500. To the nearest thousand: 44 500 to 45 499. To the nearest ten thousand: 45 000 to 54 999. Both: 45 000 to 45 499.

X3. 100. They look like 4 a b a 4: 10 choices for a and 10 for b.

X4. Largest 43 210, smallest 10 234 (the only five different digits that add up to 10 are 0, 1, 2, 3, 4).

X5. 10: 49 594, 49 694, 49 794, 49 894, 49 994, 50 005, 50 105, 50 205, 50 305, 50 405.

X6. 56. 99 990 to 99 999 is 10 numbers × 5 digits = 50, and 100 000 has 6 digits.

X7. 15. Ten thousands digit 1: the other 2 goes as 2 in one place (4 ways) or 1 and 1 in two places (6 ways): 10. Ten thousands digit 2: 4 ways. Ten thousands digit 3: 1 way (30 000). 10 + 4 + 1 = 15.

X8. Riddles will vary. Check that the rounding clue gives the right range and that only one number is left. Model riddle: "Rounded to the nearest thousand I am 30 000. All my digits are different. My digits add up to 12. I am odd." The only number is 30 261.

X9. 40 004, which is 11 km more.

Lesson 3: Operation Detectives

C1. 36 stickers. 18 + 6 × 3 = 18 + 18 = 36.

C2. 72 tickets. (18 + 6) × 3 = 24 × 3 = 72.

C3. 16 cm (56 ÷ 7 = 8, and 8 × 2 = 16).

C4. 84 children (a team has 30 ÷ 5 = 6 children; 90 − 6 = 84).

C5. 7 035

C6. 706

T1. 468 (5 units = 2 340, 1 unit = 468).

T2. (3 + 5) × 6 − 2 = 8 × 6 − 2 = 46.

T3. 12 ÷ 4 × 5 = 15 (the only way).

T4. Multiplying by 5 and then dividing by 5 gets back to your number. The 35 divided by 5 adds 7. Taking away your number leaves the 7. So the answer is always 7.

T5. 20 (18 + 7 = 25, 4 × 25 = 100, 120 − 100 = 20).

T6. 3, 7, 10, 17, 27.

T7. 21, from 3 + 4 × 5 − 2 (or 3 + 5 × 4 − 2). Put the two biggest numbers in the × and use the smallest to take away.

T8. No. 50 − 20 − 10 = 20 (left to right), but 50 − (20 − 10) = 50 − 10 = 40.

T9. 55 (425 + 15 = 440, 440 ÷ 8 = 55).

T10. (4 + 8) × (5 − 3) = 12 × 2 = 24.

T11. 618 more. 4 units = 1 236, so 1 unit = 309; chicken rice 927, laksa 309; 927 − 309 = 618.

X1. For example 1 = (4 + 4) ÷ (4 + 4); 2 = 4 ÷ 4 + 4 ÷ 4; 3 = (4 + 4 + 4) ÷ 4; 4 = 4 + (4 − 4) × 4; 5 = (4 × 4 + 4) ÷ 4. Other ways are possible.

X2. $122 (300 − 56 = 244, 244 ÷ 2 = 122; Ali has $178).

X3. For example 6 × (5 − 4 + 3) = 24 and (3 + 5 − 4) × 6 = 24. Other ways are possible.

X4. 6 (6 × 6 − 6 = 36 − 6 = 30). Try 5: 20, too small; try 7: 42, too big.

X5. 3. Work backwards: 4th = 49 − 30 = 19, 3rd = 30 − 19 = 11, 2nd = 19 − 11 = 8, 1st = 11 − 8 = 3. The chain is 3, 8, 11, 19, 30, 49.

X6. Siti 20, Ravi 60. Draw bars: Ravi 3 units, Siti 1 unit. After the gift they are equal, so Ravi's 2 extra units are 20 + 20 = 40. 1 unit = 20.

X7. 2 ways: 1 − 2 + 3 + 4 − 5 + 6 = 7 and 1 + 2 − 3 − 4 + 5 + 6 = 7. (All + makes 21, which is 14 too many. Changing a + to a − takes that number away twice, so the numbers we change must add up to 7: 2 and 5, or 3 and 4.)

X8. $48. Work backwards: $9 was half of what was left, so before the gift she had $18. Before lunch: $18 + $6 = $24. That was half, so at first she had $48.

X9. (a) Change the +35 to +50: the added number divided by 5 is what is left at the end. (b) Answers vary. Model: × 4, + 12, ÷ 4, − your number: always 3, because × 4 then ÷ 4 gets the number back, and 12 ÷ 4 = 3 is added.

Lesson 4: Fraction Frontier

C1. 3 whole pizzas and 2 slices of another pizza (11 thirds = 3 2/3 pizzas).

C2. 11 pieces (2 × 4 = 8, and 3 more). In fractions: 11 quarters = 11/4 = 2 3/4 bars.

C3. 8 counters in a group. Ali takes 1 group out of 4, which is 1/4 of the counters (1/4 of 32 = 8).

C4. 12 counters (each group has 6; 2 groups = 12). That is 2/3 of 18.

C5. The strip with 5 of 6 parts shaded (5/6). Two of the three parts is 4 of the 6 small parts (2/3 = 4/6), and 5 is more than 4.

C6. 3/10 of the ribbon is left (10 − 3 − 4 = 3 parts).

T1. Ben. 7/4 = 14/8 = 1 6/8, which is more than 1 5/8.

T2. 25 (1 unit = 5, blue = 15, not blue = 5 units = 25).

T3. 1/2, 7/12, 2/3, 5/6 (6/12, 7/12, 8/12, 10/12).

T4. 17/6 = 2 5/6. Between 2 and 3: 13/6, 14/6, 15/6, 16/6, 17/6. Odd numerators: 13/6, 15/6, 17/6. 15/6 = 2 1/2 is exactly in the middle and 13/6 is closer to 2, so only 17/6 is closer to 3.

T5. 3/10 and 4/10 (1/5 = 2/10 and 1/2 = 5/10 do not count).

T6. 12 pieces (4 in each metre).

T7. 19 (4 wholes = 16 quarters, plus 3).

T8. No. Both have 3 pieces, but sevenths are smaller pieces than fifths, so 3/5 is bigger.

T9. 40 (2 units = 16, 1 unit = 8, 5 units = 40).

X1. 12/16. 3/4 has 3 + 4 = 7 parts; 28 ÷ 7 = 4, so 3 × 4 = 12 and 4 × 4 = 16.

X2. $60. Book 4/12, pen 3/12, so 5/12 is left. 5 units = $25, 1 unit = $5, 12 units = $60.

X3. 40 litres. 3/4 = 6/8, so 15 litres is 3/8. 1/8 = 5 litres, 8/8 = 40 litres.

X4. 5/7. It is 2/7 short of 1. The next best, 2/3, is 1/3 = 2/6 short, and 2/7 is less than 2/6.

X5. 24. Red 3/12 and blue 4/12, so green is 5/12. 5 units = 10, 1 unit = 2, 12 units = 24.

X6. 24 marbles. Ali gets 4/12 and Mei gets 3/12, so Ali gets 1/12 more than Mei. 1/12 = 2 marbles, so 12/12 = 24. Check: Ali 8, Mei 6.

X7. (a) 30 sweets: 1/2 + 1/3 = 5/6 are taken, so 1/6 is left; 1/6 = 5, so 6/6 = 30. (b) 42 sweets (1/6 = 7). The jar always holds 6 times the number left. (c) Puzzles will vary: check that the fractions taken add to less than 1 and that the left-over fraction was found first.

X8. 18. Half = 3 sixths and a third = 2 sixths, so 1 sixth = 3 and the number is 6 sixths = 18. Check: 9 is 3 more than 6.

X9. 150 pages. Cut each fifth in half to make tenths: Monday 4 tenths, the rest is 6 tenths, Tuesday 3 tenths, 3 tenths left = 45. 1 tenth = 15, 10 tenths = 150.

Lesson 5: Decimal Detectives

C1. $3.47

C2. Ali has more: 90¢ is 1¢ more than 89¢.

C3. 7.5 (7.46 is just past the halfway point 7.45).

C4. $3.75

C5. 0.47 m (4 tenths and 7 hundredths).

C6. 3.6, then 3.65.

T1. 0, 1, 2 and 3 (3.45 is more than 3.42)

T2. Smallest 9.50, largest 10.49

T3. 2.65 and 2.605

T4. 4.687 (rounds to 4.7, so tenths 6 or 7; tenths 7 would make hundredths 9, 4.79..., too big; so 4.68_ and 4 + 6 + 8 + 7 = 25).

T5. $1.85 ($10 − $2.60 = $7.40, $7.40 ÷ 4 = $1.85)

T6. 0.06, 0.066, 0.6, 0.606, 0.66 (as 0.060, 0.066, 0.600, 0.606, 0.660)

T7. 1.44 m (Ben 1.53 m)

T8. 47.2 (2.8 away; the next closest is 42.7)

T9. 0.8 m each; 3.2 m

T10. 10 numbers (7.35, 7.36 ... 7.44). The number is 7.38 (7.47 also has digits adding to 18, but it rounds to 7.5).

T11. 12 numbers: 3.69, 3.96, 6.39, 6.93, 9.36, 9.63, 36.9, 39.6, 63.9, 69.3, 93.6, 96.3. Closest to 7: 6.93. Rounds to 40: 39.6.

X1. 2.459 (it is from 2.455 to 2.464; 2.45_ needs 9 to make 20; 2.46_ would need 8, but 2.468 rounds to 2.47)

X2. 0.2 kg (2 apples = 0.95 − 0.45 = 0.5 kg, 1 apple = 0.25 kg, pear = 0.45 − 0.25)

X3. Six 50¢ and four 20¢ (ten 20¢ make $2; each swap to a 50¢ adds 30¢; $1.80 more needs 6 swaps)

X4. 5.50, 5.51, 5.52, 5.53, 5.54

X5. $1.00 (3 pencils + $1.70 = $4.70, so 3 pencils cost $3.00; a pen costs $1.85)

X6. 19 (3.91 to 3.99 is 9, 4.00 to 4.09 is 10)

X7. Puzzles will vary: check by listing every number that can be made. Model: cards 3, 5, 6, 9 and target 40: the closest is 39.65 (0.35 away); the next is 39.56 (0.44 away). What if the target is 50? Then the closest is 53.69 (3.69 away); the closest below 50 is 39.65, which is 10.35 away.

X8. Ali $2.30, Ben $2.80, Siti $4.10 (take away $0.50 + $1.80 = $2.30, then $6.90 ÷ 3 = $2.30)

X9. 0.25 kg (half the water is 1.45 − 0.85 = 0.6 kg, all the water is 1.2 kg, jug = 1.45 − 1.2)

Lesson 6: Angle Turns

C1. 180° for a half turn and 90° for a quarter turn.

C2. 4 right angles make a full turn. Three right angles are 270°.

C3. S

C4. SW. She turned 3 × 45° = 135°.

C5. 30° for one gap (360 ÷ 12). 30° at 1 o'clock and 180° at 6 o'clock.

C6. 26° (90° − 64°)

T1. N (225° is 5 steps of 45°: S, SW, W, NW, N)

T2. 60° and 60°. Both are 2 spaces apart the shorter way.

T3. 210° (7 jumps of 5 minutes, and each jump is one clock space of 30°)

T4. Two ways: 45° clockwise and 135° anticlockwise; or 90° clockwise and 180°.

T5. 225° (W → NW → N → NE → E → SE: 5 steps of 45°)

T6. 6 angles (25°, 40°, 25°, 65°, 65°, 90°); the biggest is 90°.

T7. 270° (180° + 90°)

T8. 2:35 (the minute hand turns 6° every minute, and 150 ÷ 6 = 25 minutes; 2:10 + 25 minutes).

T9. 18° and 72° (5 equal parts make 90°, so one part is 18°)

X1. 75° (the minute hand points to 6; the hour hand is halfway from 3 to 4, so it is 2 and a half spaces from the 6: 30° + 30° + 15°)

X2. 10 angles (4 + 3 + 2 + 1). Each small angle is 30°, so the right angles are the 3-part angles: 2 of them.

X3. 8 turns. After 30 turns it faces W (30 = 3 × 8 + 6, and 6 steps from N is W).

X4. 65°. In 10 minutes the minute hand turns 60° and the hour hand turns 5° the same way, so the gap shrinks by 55°: 120° − 55° = 65°.

X5. She is right. 10 o'clock is the mirror image of 2 o'clock across the line from 12 to 6, so the hands make the same angle. The other pairs are 1 and 11 (30°), 3 and 9 (90°), 4 and 8 (120°), 5 and 7 (150°).

X6. 90° (Mei 225°, Ali 135°)

X7. Puzzles will vary. Model: 9:30 is 105° (the hour hand is halfway between 9 and 10, 3 and a half spaces, 3 × 30° + 15°). Another time with the same angle is 2:30.

X8. 15°, 30° and 45° (6 equal parts make 90°, so one part is 15°)

X9. 20° (90° + 90° = 180°, and 180° − 160° = 20°)

Lesson 7: Area Architects II

C1. 63 cm² (7 rows of 9 squares).

C2. 38 cm (15 + 4 + 15 + 4).

C3. 1 by 12, 2 by 6 and 3 by 4 (a 6 by 2 is the same rectangle turned round).

C4. 30 cm (the breadth is 7 cm; 8 + 7 + 8 + 7 = 30).

C5. 24 + 6 = 30 cm²

C6. Area 36 − 4 = 32 cm². The perimeter is the same: 24 cm.

T1. 1 by 30 (62 cm), 2 by 15 (34 cm), 3 by 10 (26 cm), 5 by 6 (22 cm). Smallest: 5 cm by 6 cm, 22 cm.

T2. 1 by 9 (9 cm²), 2 by 8 (16 cm²), 3 by 7 (21 cm²), 4 by 6 (24 cm²), 5 by 5 (25 cm²). Largest: the 5 cm by 5 cm square, 25 cm².

T3. Area 96 − 4 × 9 = 60 cm². Perimeter 40 cm, the same as the rectangle: each cut takes away 3 + 3 and adds 3 + 3.

T4. Area 4 + 9 + 16 = 29 cm². Perimeter 26 cm: push the steps out to a 9 cm by 4 cm rectangle.

T5. 80 m². The whole thing is 14 m by 10 m = 140 m²; take away the lawn 60 m².

T6. Smallest 12 cm (a 2 by 4 rectangle, or a 3 by 3 square with one corner missing). Largest 18 cm (for example a row of 8).

T7. 8 cm longer: 28 cm instead of 20 cm. The area stays 24 cm².

T8. 8 cm by 5 cm. (List the pairs for 40: 1 by 40, 2 by 20, 4 by 10, 5 by 8; only 5 by 8 has perimeter 26 cm.)

T9. 6 cm by 6 cm, a square, with perimeter 24 cm (the others: 1 by 36 is 74 cm, 2 by 18 is 40 cm, 3 by 12 is 30 cm, 4 by 9 is 26 cm).

X1. 64 cm². Each rectangle is one whole side and one half side all round, which is 3 sides of the square. 24 ÷ 3 = 8 cm, and 8 × 8 = 64.

X2. Area 60 − 8 = 52 cm². Perimeter 32 + 4 + 4 = 40 cm: the two 4 cm sides inside the gap cannot be pushed out.

X3. (a) 36 m², a 6 m by 6 m square. (b) 72 m²: the two sides from the wall are 6 m and the side opposite the wall is 12 m (6 + 12 + 6 = 24).

X4. Area 16 + 16 − 4 = 28 cm² (the 2 by 2 middle is counted once). Perimeter 32 cm: each of the 4 arms is 3 cm long, so each arm adds 3 + 2 + 3 = 8 cm.

X5. +1 m: 64 − 49 = 15 m². +2 m: 81 − 49 = 32 m². +3 m: 100 − 49 = 51 m². Each time the new area is two strips of 7 m by the extra width, plus a small square in the corner: 14 × 1 + 1 × 1, 14 × 2 + 2 × 2, 14 × 3 + 3 × 3. The gains 15, 32, 51 grow faster each time.

X6. Mei 6 cm by 6 cm (24 cm), Ben 2 cm by 18 cm (40 cm). The perimeters for 36 are 74, 40, 30, 26 and 24; only 40 and 24 differ by 16.

X7. Puzzles will vary: check that every whole-metre pen is listed. Model: 30 m of fence, no wall: the sides add to 15, and 7 m by 8 m gives the largest area, 56 m². With a wall on one side: 7 m, 16 m, 7 m (or 8 m, 14 m, 8 m) both give the largest area, 112 m².

X8. 63 m². Slide the paths to the edges: the grass makes a 9 m by 7 m rectangle. (Or 80 − 10 − 8 + 1 = 63, because the square where the paths cross was counted twice.)

X9. Area 36 cm², so the side is 6 cm. The perimeter went down from 26 cm to 24 cm.

Lesson 8: Symmetry and Nets

C1. 6 faces, 12 edges, 8 corners.

C2. 4 ways (across the middle both ways, and along both diagonals).

C3. 2 ways (across the middle both ways). Folding along a diagonal does not make the halves match.

C4. 5

C5. A, B, H and T. F and L cannot.

C6. No. Four squares wrap round to make the walls, but the 5th and 6th land on top of the 1st and 2nd, and two ends are left open.

T1. 22 squares: 2 faces of 3 by 2 (6 each), 2 faces of 3 by 1 (3 each) and 2 faces of 2 by 1 (2 each). 12 + 6 + 4 = 22.

T2. Bottom 4 (3 + 4 = 7). Back 2 (5 + 2 = 7).

T3. 2. All six faces add up to 1 + 2 + 3 + 4 + 5 + 6 = 21, and 21 − 19 = 2.

T4. Only the T. The line has 2, the square has 4, the L and the zigzag have 0.

T5. No. It has a 2 by 2 block of squares: 4 squares meet at one corner, but on a cube only 3 faces meet at a corner, so two of them would overlap.

T6. Only the row of 5 squares (one line up-down through the middle square and one line along the row).

T7. 4 faces touch it. 5 cannot: the face opposite red never touches it.

T8. Answers vary. One line: for example the whole top row (3 squares), or the top-left, top-right and centre squares. Two lines: the top-middle and bottom-middle squares, or the left-middle and right-middle squares, or two opposite corners.

X1. 3. Two dice have 21 + 21 = 42 altogether. The hidden faces add up to 42 − 36 = 6, and they show the same number, so each is 3.

X2. With 5 in a row, four squares wrap round to make a ring of walls and the 5th lands on top of the 1st, so two squares share a face and no face is left for the other end. The most is 4 in a row (for example the cross-shaped net).

X3. 8. Only 4 shapes do not fold: the row of 5, the V (an L with arms of 3 and 3), the U (a C shape) and the P (a 2 by 2 block with one extra square).

X4. 16. The top dice: its bottom is 2 (5 + 2 = 7). The middle dice: top and bottom are opposite, 7. The bottom dice: top and bottom, 7. 2 + 7 + 7 = 16.

X5. Yes. The 4 side faces are two opposite pairs, and each pair adds up to 7: 7 + 7 = 14.

X6. Opposite 1 is 6 (the only face it does not touch). Opposite 2 is 4. That leaves 3 and 5, so opposite 3 is 5.

X7. Designs will vary: check each pair of opposite faces on the net adds up to 7. Model: a row of four squares 2, 1, 5, 6 with a 3 above the 1 and a 4 below the 1. Opposite pairs: 2 and 5, 1 and 6 (first and third, second and fourth in the row), and 3 and 4 (above and below).

X8. 6. Both squares in the middle column: 3 ways. One square in each side column, in the same row: 3 ways.

X9. The middle square and the 4 corners (an X), or the middle square and the 4 edge-middle squares (a plus).

Lesson 9: Data Detectives

C1. 35

C2. 15 bowls

C3. 100

C4. 10 am

C5. 68 (Mei has 43; 25 + 43 = 68)

C6. 35

T1. Neither: both have 34 girls. (4A: 28 + 6 = 34. 4B: 63 − 5 = 58, 58 ÷ 2 = 29 boys, 29 + 5 = 34 girls.)

T2. 100 (40 + 20 + 20 + 20)

T3. 65 (300 − 65 − 80 − 90)

T4. From age 7 to 8 (7 cm). Altogether 17 cm.

T5. Week 4 (45 is more than 44).

T6. No. 60 is only 10 more than 50. It looks twice as high because 50 is 10 above the bottom (40) and 60 is 20 above it.

T7. Tuesday to Wednesday: an increase of 45. (The other changes are down 25 and down 30.)

T8. 23 °C (5 hours, so 10 °C less)

X1. Week 1: 35 (4 jumps make 40, so 10 a week). First more than 100: week 8 (105).

X2. Tuesday 50, Wednesday 50, Thursday 60. (Tue + Wed + Thu = 160; take away the extra 10, 150 shared by 3 = 50.)

X3. Mei's: hers went up 30 and Ben's went up 6, so 24 more.

X4. No. Monday is 300 and Tuesday is 500. 3 times 300 is 900.

X5. 40. Work backwards: 12 pm 60, 11 am 70, 10 am 55, 9 am 40.

X6. 60 (60, 70, 80, 90, 100)

X7. A 34, B 17, C 27, D 22. (A + B + C = 78: four equal parts and 10 more, so one part is 68 ÷ 4 = 17.)

X8. After 4 hours: 56 litres each. The gap of 60 litres closes by 15 litres every hour.

X9. Answers vary. (a) Start at 50 with each line worth 1: the line climbs steeply. (b) Start at 0 with each line worth 20: the line is almost flat. A fair graph starts at 0 or says clearly that it does not, so the viewer can see the real change of 6. Both graphs show the same numbers.

Lesson 10: Challenge Day

C1. 9 and 10

C2. 3 rabbits (and 5 chickens). Suppose all chickens: 16 legs, 6 short, each swap adds 2.

C3. 20 (70 ÷ 2 = 35, 35 − 15 = 20)

C4. 11 (5 + 1 + 5)

C5. 12 and 18

C6. 2 (and four 10¢ coins: 100¢ + 40¢ = 140¢)

T1. 7 tricycles (and 13 bicycles). All bicycles: 40 wheels, 7 short, each swap adds 1.

T2. 24 and 26

T3. $51 (18 × 2 = 36, 36 + 15 = 51)

T4. Ben 11, Siti 10, Ali 9.

T5. 22 (8 + 1 + 13)

T6. 5 spiders (and 7 ants). All ants: 72 legs, 10 short, each swap adds 2.

T7. 40 (38 + 39 + 40)

T8. 6 (in 6 years the total is 60: 5 equal bars, so Siti will be 12)

X1. 19 (7, 17, 27, 37, 47, 57, 67, 87, 97 and 70 to 79; 77 is counted once)

X2. 14 chickens (28 legs) and 6 goats (24 legs).

X3. 47 (74 – 47 = 27)

X4. Seat 1 Ali, seat 2 Siti, seat 3 Ben, seat 4 Mei.

X5. $4 (a cake is $7). Suppose all 7 are drinks: take away 4 × $3 = $12, then $28 ÷ 7 = $4.

X6. 5 (uncle 20; in 10 years 15 and 30)

X7. 8 tables seat 6 (and 7 seat 4). All seat 4: 60 seats, 16 short, each swap adds 2.

X8. 32. Work backwards: (5 + 2) × 2 = 14, (14 + 2) × 2 = 32.

X9. Puzzles will vary: check by trying every possible number. Model: 10 vehicles, 32 wheels, cars (4 wheels) and motorcycles (2 wheels): 6 cars and 4 motorcycles is the only way. Why 35 is impossible: every chicken has 2 legs and every goat has 4, so the total number of legs is always even, and 35 is odd.